

AI Agent Red Team Simulator

Authorized, local-first security testing for AI agents - from prompt behavior to tools, services, and auditable evidence.

Status: Active	Date: July 2026	Project Lead: Vishal	Quick Links: GitHub Repo
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1. The Big Picture

The Problem

AI agents combine probabilistic model behavior with prompts, tools, credentials, APIs, memory, and deployment infrastructure. Traditional scanners rarely test whether manipulated instructions can expose sensitive context, bypass policy, or trigger unsafe capabilities.

The Solution

The simulator provides one assessment fabric for explicit Python targets, live HTTP agents, authorized hosted endpoints, and isolated Kali workflows. Curated and adaptive probes feed a common evaluator that records severity, evidence, redacted responses, timelines, and remediation-oriented reports.

Key Impact Metric: 47 automated tests passed, six explicitly enrolled agents discovered, and a deterministic smoke assessment completed with 6 PASS / 0 FAIL / 0 ERROR.

2. Core Features & Capabilities

- Multi-path assessment:** Curated payloads, local adaptive probes, live HTTP discovery, and bounded Kali testing cover both model and application behavior.
- Fail-closed evaluation:** PASS, FAIL, ERROR, and UNPARSED outcomes include severity, detector evidence, secret redaction, and automation-friendly exit behavior.
- Enterprise evidence:** JSON, Markdown, timelines, JSONL events, and remediation guidance support release gates, triage, regression, and audit review.

3. Technical Implementation

Layer	Technology Selection	Strategic Rationale
Orchestration	Python CLI, natural-language assistant	Interprets intent, scopes tests, discovers targets, and records observable assessment phases.
AI Runtime	Ollama, LangGraph, local agents	Keeps adaptive generation local while supporting tool-using weather and travel target services.
Security Lab	Kali, SSH tunnels, HTTP probes	Runs bounded recon and agent-aware tests without exposing loopback services to the LAN.

Layer	Technology Selection	Strategic Rationale
Evidence	JSON, Markdown, JSONL	Normalizes findings, redacts configured secrets, and produces reviewable reports and event traces.

4. Key Results & Visual Assets

Measurable Metrics

- **Validation:** 47 of 47 unit tests passed and the reviewed core modules compiled cleanly.
- **Coverage:** Six explicit targets were discovered; the prompt-disclosure smoke scan returned 6 PASS, 0 FAIL, 0 ERROR.

Assessment Architecture



Figure 1: Authorized assessment flow from operator intent to auditable evidence.